

Question 1 (20 Points)

(4 P) a) **Formulation of the transportation problem as linear optimization problem in standard form**

- Optimization variables x_i : Transported quantity
- Cost function: transportation cost $\sum_{i=1}^4 C_i x_i$
- Equality constraints: Constant demand D_i , $i = 1, 2$
- Inequality constraints: No negative transport quantities

$\Rightarrow$ Optimization problem:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^4} \quad & C_1 x_1 + C_2 x_2 + C_3 x_3 + C_4 x_4 \\ \text{s.t.} \quad & x_1 + x_2 - x_4 = D_1 \\ & x_3 + x_4 = D_2 \\ & x_i \geq 0, \quad i = 1, \dots, 4 \end{aligned}$$

The Problem is already in standard form

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

$$\text{with } \mathbf{c}^T = [1, 2, 10, 0.1], \mathbf{b} = \begin{bmatrix} 20 \\ 10 \end{bmatrix} \text{ and } \mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

(4 P) b) **Formulation of the KKT conditions**

Lagrangian function:

$$\begin{aligned} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) &= \mathbf{c}^T \mathbf{x} + \boldsymbol{\lambda}^T (\mathbf{A} \mathbf{x} - \mathbf{b}) + \boldsymbol{\mu}^T (-\mathbf{x}) \\ &= \sum_{i=1}^4 C_i x_i + \lambda_1 (x_1 + x_2 - x_4 - D_1) + \lambda_2 (x_3 + x_4 - D_2) - \sum_{i=1}^4 \mu_i x_i \end{aligned}$$

KKT conditions:

$$\begin{aligned} \nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) &= \begin{bmatrix} 1 + \lambda_1 - \mu_1 \\ 2 + \lambda_1 - \mu_2 \\ 10 + \lambda_2 - \mu_3 \\ 0.1 - \lambda_1 + \lambda_2 - \mu_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\ \nabla_{\boldsymbol{\lambda}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) &= \begin{bmatrix} x_1 + x_2 - x_4 - D_1 \\ x_3 + x_4 - D_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ x_i &\geq 0, \quad i = 1, \dots, 4 \\ \mu_i &\geq 0, \quad i = 1, \dots, 4 \\ x_i \mu_i &= 0, \quad i = 1, \dots, 4 \end{aligned}$$

(4P) c) Check for basic feasible solution

Basis $\mathcal{B}$	Basis matrix $\mathbf{B}$	$\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b}$	$\mathbf{x}_N = \mathbf{0}$	basic feasible solution?
$\mathcal{B}_1 = \{1,2\}$	$\mathbf{B}_1 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$	-	-	no, since $\mathbf{B}_1$ singular
$\mathcal{B}_2 = \{1,3\}$	$\mathbf{B}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$x_1 = 20 > 0$ $x_3 = 10 > 0$	$x_2 = 0$ $x_4 = 0$	yes
$\mathcal{B}_3 = \{3,4\}$	$\mathbf{B}_3 = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$	$x_3 = 30 > 0$ $x_4 = -10 < 0$	$x_1 = 0$ $x_2 = 0$	no, since $x_{B,2} < 0$

 (8P) d) Solution with Simplex method and initial basis $\mathcal{B}_2 = \{1,3\}$

$$\mathcal{B} = \{1,3\}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 20 \\ 10 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} 1 \\ 10 \end{bmatrix}$$

$$\mathcal{N} = \{2,4\}, \quad \mathbf{N} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} 2 \\ 0.1 \end{bmatrix}$$

Evaluation of the KKT conditions

$$\boldsymbol{\lambda} = -(\mathbf{B}^T)^{-1}\mathbf{c}_B = -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 10 \end{bmatrix} = \begin{bmatrix} -1 \\ -10 \end{bmatrix}$$

$$\boldsymbol{\mu}_N = \mathbf{c}_N + \mathbf{N}^T\boldsymbol{\lambda} = \begin{bmatrix} 2 \\ 0.1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -10 \end{bmatrix} = \begin{bmatrix} 1 \\ -8.9 \end{bmatrix}$$

shows, that this is not the minimum ($\mu_{N,2} = \mu_4 < 0$). One step of the Simplex method leads to

$$s = 4, \quad \mathbf{d} = \mathbf{B}^{-1}\mathbf{A}_4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$r = 3, \text{ (2. basis element), since only } d_2 > 0, \quad x_s^+ = \frac{x_{B,2}}{d_2} = 10$$

$$\mathbf{x}_B^+ = \mathbf{x}_B - \mathbf{d}x_s^+ = \begin{bmatrix} 30 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1^+ \\ x_3^+ \end{bmatrix}, \quad \mathbf{x}_N^+ = \begin{bmatrix} 0 \\ x_s^+ \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \end{bmatrix} = \begin{bmatrix} x_2^+ \\ x_4^+ \end{bmatrix}.$$

For the next Simplex step the elements from $\mathcal{B}$ and $\mathcal{N}$ with the indices $r \in \mathcal{B}$ and $s \in \mathcal{N}$ are swapped, resulting in the *new feasible solution*

$$\mathcal{B} = \{1,4\}, \quad \mathbf{B} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 30 \\ 10 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}$$

$$\mathcal{N} = \{2,3\}, \quad \mathbf{N} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} 2 \\ 10 \end{bmatrix}.$$

Evaluation of the KKT conditions for the new point results in

$$\boldsymbol{\lambda} = -(\mathbf{B}^T)^{-1}\mathbf{c}_B = -\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1.1 \end{bmatrix}$$

$$\boldsymbol{\mu}_N = \mathbf{c}_N + \mathbf{N}^T\boldsymbol{\lambda} = \begin{bmatrix} 2 \\ 10 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -1.1 \end{bmatrix} = \begin{bmatrix} 1 \\ 8.9 \end{bmatrix}.$$

Since both Lagrangian multipliers $\boldsymbol{\mu}_N = [\mu_{N,1}, \mu_{N,2}]^T = [\mu_2, \mu_3]^T$ are positive, the point $\mathbf{x}^* = [30, 0, 0, 10]^T$ is the *optimal solution*.

Question 2 (25 Points)

(9P) a) Lagrangian function and linear quadratic approximation of the nonlinear problem

Lagrangian function:

$$\begin{aligned} L(\mathbf{x}, \boldsymbol{\mu}) &= f(\mathbf{x}) + \boldsymbol{\mu}^T \mathbf{h}(\mathbf{x}) \\ &= k_1 x_1^3 + k_2 x_1 x_2 + k_3 x_2^5 + \mu_1 \left(-\frac{1}{2} x_1 x_2 - \frac{1}{2} x_2 + \frac{7}{8} \right) + \mu_2 \left(-\frac{1}{2} x_1 + x_2 - \frac{5}{8} \right) + \mu_3 \left(\frac{1}{2} x_1^3 x_2 - \frac{7}{2} \right) \end{aligned}$$

Partial derivative of the Lagrangian:

$$\begin{aligned} \nabla_{x_1} L(\mathbf{x}, \boldsymbol{\mu}) &= 3k_1 x_1^2 + k_2 x_2 - \frac{1}{2} \mu_1 x_2 - \frac{1}{2} \mu_2 + \frac{3}{2} \mu_3 x_1^2 x_2 \\ \nabla_{x_2} L(\mathbf{x}, \boldsymbol{\mu}) &= k_2 x_1 + 5k_3 x_2^4 - \frac{1}{2} \mu_1 x_1 - \frac{1}{2} \mu_1 + \mu_2 + \frac{1}{2} \mu_3 x_1^3 \\ \nabla_{x_1 x_1} L(\mathbf{x}, \boldsymbol{\mu}) &= 6k_1 x_1 + 3\mu_3 x_1 x_2 \\ \nabla_{x_1 x_2} L(\mathbf{x}, \boldsymbol{\mu}) &= \nabla_{x_2 x_1} L(\mathbf{x}, \boldsymbol{\mu}) = k_2 - \frac{1}{2} \mu_1 + \frac{3}{2} \mu_3 x_1^2 \\ \nabla_{x_2 x_2} L(\mathbf{x}, \boldsymbol{\mu}) &= 20k_3 x_2^3 \end{aligned}$$

Plug-in current iteration point:

$$\nabla_{xx}^2 L(\mathbf{x}^k, \boldsymbol{\mu}^k) = \begin{bmatrix} 6k_1 & k_2 \\ k_2 & 20k_3 \end{bmatrix}, \quad \nabla f(\mathbf{x}^k) = \begin{bmatrix} 3k_1 + k_2 \\ k_2 + 5k_3 \end{bmatrix}$$

Linearization of the constraints:

$$\begin{aligned} \nabla h_1(\mathbf{x}^k) &= \left[\begin{array}{c} -\frac{1}{2} x_2 \\ -\frac{1}{2} x_1 - \frac{1}{2} \end{array} \right]_{\mathbf{x}=\mathbf{x}^k} = \left[\begin{array}{c} -\frac{1}{2} \\ -1 \end{array} \right], \quad h_1(\mathbf{x}^k) = -\frac{1}{8} \\ \nabla h_2(\mathbf{x}^k) &= \left[\begin{array}{c} -\frac{1}{2} \\ 1 \end{array} \right], \quad h_2(\mathbf{x}^k) = -\frac{1}{8} \\ \nabla h_3(\mathbf{x}^k) &= \left[\begin{array}{c} \frac{3}{2} x_1^2 x_2 \\ \frac{1}{2} x_1^3 \end{array} \right]_{\mathbf{x}=\mathbf{x}^k} = \left[\begin{array}{c} \frac{3}{2} \\ \frac{1}{2} \end{array} \right], \quad h_3(\mathbf{x}^k) = -3 \end{aligned}$$

$\Rightarrow$ LQ approximation:

$$\begin{aligned} \min_{\mathbf{s} \in \mathbb{R}^2} & \frac{1}{2} \mathbf{s}^T \begin{bmatrix} 6k_1 & k_2 \\ k_2 & 20k_3 \end{bmatrix} \mathbf{s} + \begin{bmatrix} 3k_1 + k_2 \\ k_2 + 5k_3 \end{bmatrix}^T \mathbf{s} \\ \text{s.t.} & \quad -\frac{1}{2} s_1 - s_2 - \frac{1}{8} \leq 0 \\ & \quad -\frac{1}{2} s_1 + s_2 - \frac{1}{8} \leq 0 \\ & \quad \frac{3}{2} s_1 + \frac{1}{2} s_2 - 3 \leq 0 \end{aligned}$$

(9P) b) **Graphical solution of the approximation**

$$\mathbf{s}^* = \mathbf{s}^k = \left[-\frac{1}{4}, 0\right]^T$$

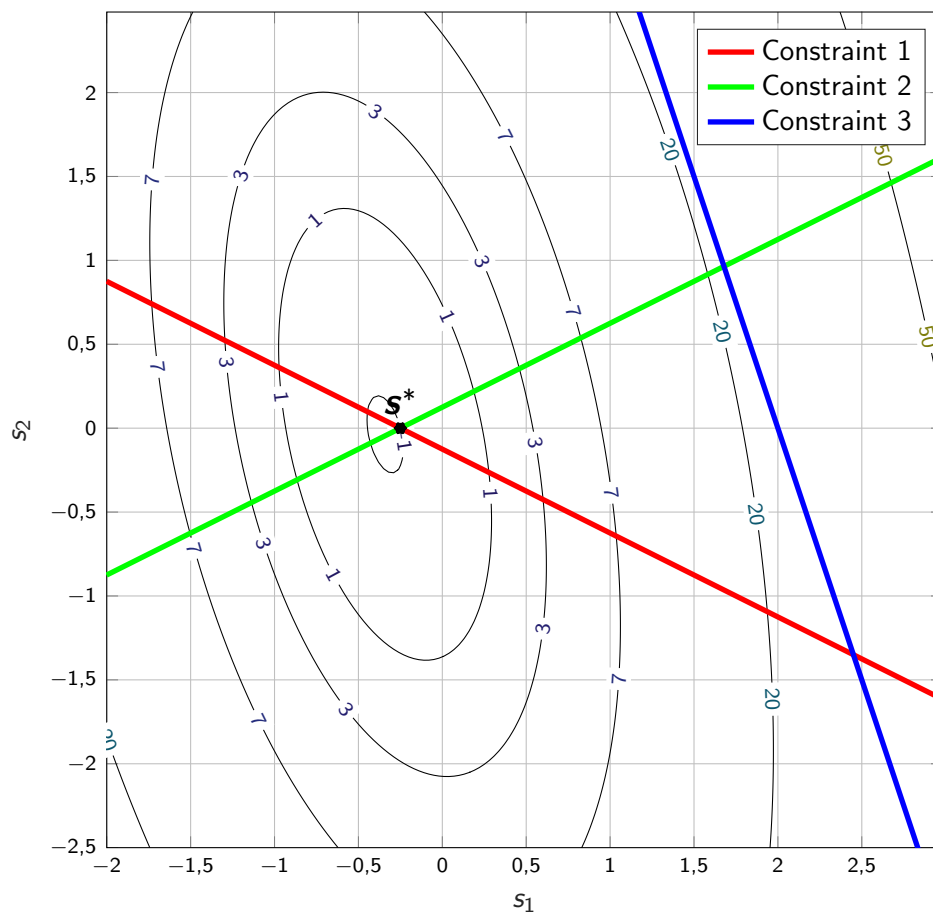


Figure 1: Graphical solution of the linear quadratic approximation.

Calculation of the new iteration point with full step size $\alpha = 1$:

$$\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha^k \mathbf{s}^k \stackrel{\alpha=1}{=} \left[\frac{3}{4}, 1\right]^T$$

Check nonlinear constraints at new iteration point (i.e. check if the full step size $\alpha = 1$ is allowed, or if one of the nonlinear constraints is violated at the new iteration point):

$$h_1(\mathbf{x}^{k+1}) = -\frac{1}{24} - \frac{1}{2} + \frac{7}{8} = -\frac{3}{8} - \frac{4}{8} + \frac{7}{8} = 0$$

$$h_2(\mathbf{x}^{k+1}) = -\frac{1}{24} + 1 - \frac{5}{8} = 0$$

$$h_3(\mathbf{x}^{k+1}) = \frac{1}{2} \left(\frac{3}{4}\right)^3 - \frac{7}{2} < 0$$

$\Rightarrow$ Full step is possible.

(7P) c) **Derivation of the interior point method for quadratic problems**

- Set up KKT conditions for QP:

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \frac{1}{2} \mathbf{x}^\top \mathbf{G} \mathbf{x} + \boldsymbol{\lambda}^\top (\mathbf{A} \mathbf{x} - \mathbf{b}) + \boldsymbol{\mu}^\top (-\mathbf{x})$$

$$\nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \mathbf{G} \mathbf{x} + \mathbf{c} + \mathbf{A}^\top \boldsymbol{\lambda} - \boldsymbol{\mu} = \mathbf{0}$$

$$\mathbf{A} \mathbf{x} - \mathbf{b} = \mathbf{0}$$

$$\mathbf{x} \geq \mathbf{0}$$

$$\boldsymbol{\mu} \geq \mathbf{0}$$

$$\mu_i x_i = 0 \quad i = 1, \dots, n$$

- Relaxation of the complementary condition $\mu_i x_i = \tau$ yields the nonlinear system of equations

$$\mathbf{F}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \begin{bmatrix} \mathbf{G} \mathbf{x} + \mathbf{c} + \mathbf{A}^\top \boldsymbol{\lambda} - \boldsymbol{\mu} \\ \mathbf{A} \mathbf{x} - \mathbf{b} \\ \mathbf{M} \mathbf{X} \mathbf{e} \end{bmatrix} - \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \tau \mathbf{e} \end{bmatrix} = \mathbf{0}, \quad \mathbf{x} \geq \mathbf{0}, \boldsymbol{\mu} \geq \mathbf{0},$$

with $\mathbf{M} = \text{diag}(\boldsymbol{\mu})$, $\mathbf{X} = \text{diag}(\mathbf{x})$ and $\mathbf{e} = [1, \dots, 1]^\top$ which can be solved with the Newton's method

$$\begin{bmatrix} \mathbf{x}^{k+1} \\ \boldsymbol{\lambda}^{k+1} \\ \boldsymbol{\mu}^{k+1} \end{bmatrix} = \begin{bmatrix} \mathbf{x}^k \\ \boldsymbol{\lambda}^k \\ \boldsymbol{\mu}^k \end{bmatrix} + \alpha^k \mathbf{s}^k \quad \text{with} \quad \mathbf{s}^k = \begin{bmatrix} \Delta \mathbf{x}^k \\ \Delta \boldsymbol{\lambda}^k \\ \Delta \boldsymbol{\mu}^k \end{bmatrix}.$$

based on

$$\underbrace{\begin{bmatrix} \mathbf{G} & \mathbf{A}^\top & -\mathbf{I} \\ \mathbf{A} & \mathbf{0} & \mathbf{0} \\ \mathbf{M}^k & \mathbf{0} & \mathbf{X}^k \end{bmatrix}}_{\nabla F(\mathbf{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)} \begin{bmatrix} \Delta \mathbf{x}^k \\ \Delta \boldsymbol{\lambda}^k \\ \Delta \boldsymbol{\mu}^k \end{bmatrix} = \underbrace{\begin{bmatrix} -\mathbf{G} \mathbf{x}^k - \mathbf{c} - \mathbf{A}^\top \boldsymbol{\lambda}^k + \boldsymbol{\mu}^k \\ -\mathbf{A} \mathbf{x}^k + \mathbf{b} \\ -\mathbf{M}^k \mathbf{X}^k + \tau^k \mathbf{e} \end{bmatrix}}_{-\mathbf{F}(\mathbf{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)}.$$

- Update of the barrier parameter τ^k with the duality measure $\bar{\mu} = \frac{1}{n} \sum_{i=1}^n \mu_i^{k+1} x_i^{k+1}$:

$$\tau^{k+1} = \sigma^{k+1} \bar{\mu}$$

with reduction factor $\sigma^{k+1} \in [0, 1]$.

- When determining the step size α^k the inequality constraints $\mathbf{x} \geq \mathbf{0}, \boldsymbol{\mu} \geq \mathbf{0}$ must be satisfied in each iteration by using a suitable line search.

Question 3 (30 Points)(3 P) a) **Hamiltonian function and necessary first-order optimality conditions**

$$H(x, u, \lambda) = l(x, u) + \lambda^T f(x, u) = \frac{1}{2}qx^2 + \frac{1}{2}ru^2 + \lambda(ax + bu)$$

Necessary first-order optimality conditions:

$$\dot{x} = \frac{\partial H}{\partial \lambda} = ax + bu, \quad x(0) = x_0 \quad (4a)$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -qx - a\lambda, \quad \lambda(T) = 0 \quad (4b)$$

$$0 = \frac{\partial H}{\partial u} = ru + b\lambda \quad (4c)$$

(10 P) b) **Determination of the optimal trajectories from the optimality conditions**

From (4c) follows

$$u = -\frac{b}{r}\lambda \Rightarrow \dot{x} = ax - \frac{b^2}{r}\lambda. \quad (5)$$

- Calculation of the optimal trajectory for state x: Derivative of $\dot{x}$

$$\begin{aligned} \ddot{x} &= a\dot{x} - \frac{b^2}{r}\dot{\lambda} \\ &= a^2x - \frac{ab^2}{r}\lambda + \frac{qb^2}{r}x + \frac{ab^2}{r}\lambda \\ &= (a^2 + \frac{qb^2}{r})x \end{aligned}$$

Define $s = a^2 + \frac{qb^2}{r} > 0$ for $q, r > 0$, i.e. $\ddot{x} = sx$

$$\begin{aligned} \Rightarrow x(t) &= c_1 e^{\sqrt{s}t} + c_2 e^{-\sqrt{s}t} = e^{\sqrt{s}t}(c_1 + c_2 e^{-2\sqrt{s}t}) \\ \dot{x}(t) &= c_1 \sqrt{s} e^{\sqrt{s}t} - c_2 \sqrt{s} e^{-\sqrt{s}t} \end{aligned}$$

- Calculation of the optimal trajectory for the adjoint state: From (5) follows

$$\begin{aligned} \lambda &= \frac{r}{b^2}(ax - \dot{x}) \\ &= \frac{r}{b^2}(ac_1 e^{\sqrt{s}t} + ac_2 e^{-\sqrt{s}t} - c_1 \sqrt{s} e^{\sqrt{s}t} + c_2 \sqrt{s} e^{-\sqrt{s}t}) \\ &= \frac{r}{b^2}[(a - \sqrt{s})c_1 e^{\sqrt{s}t} + (a + \sqrt{s})c_2 e^{-\sqrt{s}t}] \\ &= \frac{r}{b^2}e^{\sqrt{s}t}[(a - \sqrt{s})c_1 + (a + \sqrt{s})c_2 e^{-2\sqrt{s}t}] \end{aligned}$$

From that follows the optimal input trajectory

$$u = -\frac{b}{r}\lambda = -\frac{1}{b}e^{\sqrt{s}t}[(a - \sqrt{s})c_1 + (a + \sqrt{s})c_2 e^{-2\sqrt{s}t}].$$

- Calculation of the constants c_1 and c_2 from the boundary constraints:

$$x(0) = x_0 = c_1 + c_2 \Rightarrow c_2 = x_0 - c_1 \quad (6)$$

$$\lambda(T) = 0 = \frac{r}{b^2} [(a - \sqrt{s})c_1 e^{\sqrt{s}T} + (a + \sqrt{s})c_2 e^{-\sqrt{s}T}] \quad (7)$$

Substitution of (6) into (7):

$$\begin{aligned} 0 &= \frac{r}{b^2} [(a - \sqrt{s})c_1 e^{\sqrt{s}T} + (a + \sqrt{s})(x_0 - c_1)e^{-\sqrt{s}T}] \\ &= c_1 [(a - \sqrt{s})e^{\sqrt{s}T} - (a + \sqrt{s})e^{-\sqrt{s}T}] + x_0(a + \sqrt{s})e^{-\sqrt{s}T} \\ \Rightarrow c_1 &= x_0 \frac{(a + \sqrt{s})e^{-\sqrt{s}T}}{-(a - \sqrt{s})e^{\sqrt{s}T} + (a + \sqrt{s})e^{-\sqrt{s}T}} \\ &= x_0 \frac{(a + \sqrt{s})}{-(a - \sqrt{s})e^{2\sqrt{s}T} + (a + \sqrt{s})} \\ &= x_0 \frac{1}{-\frac{(a - \sqrt{s})}{(a + \sqrt{s})}e^{2\sqrt{s}T} + 1} \\ \text{with } h &= \frac{(a - \sqrt{s})}{(a + \sqrt{s})} \Rightarrow c_1 = x_0 \frac{1}{-he^{2\sqrt{s}T} + 1} \\ \Rightarrow c_2 &= x_0 - c_1 = x_0 \left(1 - \frac{1}{-he^{2\sqrt{s}T} + 1}\right) \\ &= x_0 \frac{-he^{2\sqrt{s}T}}{-he^{2\sqrt{s}T} + 1} \end{aligned}$$

(10 P) c) Requirement for the horizon length T

Solution trajectory in dependence of T :

$$x(t) = x_0 \left(\frac{1}{-he^{2\sqrt{s}T} + 1} e^{\sqrt{s}t} + \frac{-he^{2\sqrt{s}T}}{-he^{2\sqrt{s}T} + 1} e^{-\sqrt{s}t} \right)$$

For $\|x(\Delta t)\| < \|x_0\|$ to be satisfied, it must apply that

$$\begin{aligned} \frac{1}{-he^{2\sqrt{s}T} + 1} e^{\sqrt{s}\Delta t} + \frac{-he^{2\sqrt{s}T}}{-he^{2\sqrt{s}T} + 1} e^{-\sqrt{s}\Delta t} &< 1 \\ e^{\sqrt{s}\Delta t} - he^{2\sqrt{s}T} e^{-\sqrt{s}\Delta t} - (1 - he^{2\sqrt{s}T}) &< 0 \\ e^{\sqrt{s}\Delta t} - 1 + (1 - e^{-\sqrt{s}\Delta t})he^{2\sqrt{s}T} &< 0 \\ e^{\sqrt{s}\Delta t} - 1 &< (e^{-\sqrt{s}\Delta t} - 1)he^{2\sqrt{s}T} \\ \frac{e^{\sqrt{s}\Delta t} - 1}{(e^{-\sqrt{s}\Delta t} - 1)h} &< e^{2\sqrt{s}T} \\ \log\left[\frac{e^{\sqrt{s}\Delta t} - 1}{(e^{-\sqrt{s}\Delta t} - 1)}\right] &< 2\sqrt{s}T \\ \Rightarrow T &> \frac{1}{2\sqrt{s}} \log\left[\frac{e^{\sqrt{s}\Delta t} - 1}{(e^{-\sqrt{s}\Delta t} - 1)}\right] \end{aligned}$$

This corresponds to the lower bound for the MPC horizon length so that the condition $\|x(\Delta t)\| < \|x_0\|$ is satisfied for all k .

(4P) d) **Calculation of the terminal cost weighting with the algebraic Riccati equation**

For the given scalar case, the equation results in

$$\begin{aligned}
 0 &= pa + ap - pb\frac{1}{r}bp + q \\
 &= -p^2\frac{b^2}{r} + 2pa + q \\
 \Rightarrow p_{1,2} &= \frac{-2a \pm \sqrt{4a^2 + 4\frac{b^2}{r}q}}{2(-\frac{b^2}{r})} \\
 &= \frac{-2a \pm \sqrt{4(a^2 + \frac{b^2}{r}q)}}{2(-\frac{b^2}{r})} \\
 &= \frac{-ra \pm \sqrt{a^2 + \frac{b^2}{r}q}}{-b^2} \\
 &= \frac{-ra \pm r\sqrt{a^2 + \frac{b^2}{r}q}}{-b^2} \\
 &= \frac{r}{b^2}(a \pm \sqrt{a^2 + \frac{b^2}{r}q})
 \end{aligned}$$

Choose the positive solution: $p = \frac{r}{b^2}(a + \sqrt{a^2 + \frac{b^2}{r}q})$.

- Advantage: shorter horizon, problem is numerically easier to solve.
- Disadvantage: The terminal cost $\frac{1}{2}p x(T)^2$ is relatively large compared to the integration term, which distorts the desired target behavior.

 (3P) e) **Procedure for input constraints**

Pontryagin's maximum principle:

$$\begin{aligned}
 u^* &= \arg \min_{u \in [u^-, u^+]} \left(\frac{1}{2}qx^2 + \frac{1}{2}ru^2 + \lambda(ax + bu) \right) \\
 &= \arg \min_{u \in [u^-, u^+]} \frac{1}{2}ru^2 + \lambda bu
 \end{aligned}$$

Unconstrained minimum from $\frac{\partial H}{\partial u} = ru + \lambda b \stackrel{!}{=} 0$:

$$u^0 = -\frac{b}{r}\lambda$$

Optimal input:

$$u^* = \Psi(x, \lambda) = \begin{cases} u^-, & \text{if } u^0 < u^- \\ u^0, & \text{if } u^0 \in [u^-, u^+] \\ u^+, & \text{if } u^0 > u^+ \end{cases}$$

Question 4 (15 Points)

(2 P) * a) **Convergence of the quasi-Newton method**

The problem must be quadratic of the form $\min_{\mathbf{x} \in \mathbb{R}^n} \mathbf{x}^T \mathbf{G} \mathbf{x} + \mathbf{c}^T \mathbf{x}$ with positive definite matrix $\mathbf{Q}$ and the underlying line search method to determine the step size α must be exact.

(4 P) * b) **Determination of the step size for SQP**

SQP methods linearize the nonlinear constraints, i.e., for a full step with $\alpha = 1$, it cannot be guaranteed that all active constraints remain active and none of the inactive constraints are violated. In contrast, with Active set methods for QPs, we know that active constraints remain active since they are linear.

The l_1 -penalty function is called an exact penalty function because for $\sigma > \max(|\lambda^*|, \mu^*)$ it is guaranteed that $\mathbf{x}^*$ is also minimum of $P(\mathbf{x}, \sigma)$. This property is not given for l_2 -penalty functions, for example.

(4 P) * c) **Singular case**

This is called singular case. There are two ways to calculate the optimal control in this case.

- Option 1: Extend the integral cost by a so-called regularization term δu^2 with a small factor δ making $\frac{\partial H}{\partial u}$ dependent on u .
- Option 2: Calculate the time derivatives of the Hamiltonian function $\frac{dr}{dt} \frac{\partial H}{\partial u} = 0$ until the control input u occurs explicitly. The derivative at which this is the case is called the degree of singularity r .

(5 P) * d) **Full discretization**

- Discretized optimal control problem

$$\hat{\mathbf{u}} = [\mathbf{u}^0, \dots, \mathbf{u}^N]^T \in \mathbb{R}^{m(N+1)}, \quad \hat{\mathbf{x}} = [\mathbf{x}^0, \dots, \mathbf{x}^N]^T \in \mathbb{R}^{n(N+1)}.$$

$$\min_{\hat{\mathbf{x}}, \hat{\mathbf{u}}} \hat{J}(\hat{\mathbf{x}}, \hat{\mathbf{u}}) = \frac{1}{2} (\mathbf{x}^N)^T \mathbf{S} \mathbf{x}^N + \sum_{k=0}^{N-1} \frac{t^{k+1} - t^k}{2} \left[(\mathbf{x}^k)^T \mathbf{Q} \mathbf{x}^k + (\mathbf{u}^k)^T \mathbf{R} \mathbf{u}^k + (\mathbf{x}^{k+1})^T \mathbf{Q} \mathbf{x}^{k+1} + (\mathbf{u}^{k+1})^T \mathbf{R} \mathbf{u}^{k+1} \right]$$

$$\text{s.t.} \quad \frac{\mathbf{x}^{k+1} - \mathbf{x}^k}{t^{k+1} - t^k} = \frac{1}{2} \left[\mathbf{A} \mathbf{x}^k + \mathbf{B} \mathbf{u}^k + \mathbf{A} \mathbf{x}^{k+1} + \mathbf{B} \mathbf{u}^{k+1} \right], \quad k = 0, 1, \dots, N-1$$

$$\mathbf{x}^0 = \mathbf{x}_0,$$

$$\mathbf{C} \mathbf{x}^k \leq \mathbf{d}, \quad k = 0, 1, \dots, N$$

$$\mathbf{u}^k \in [\mathbf{u}^-, \mathbf{u}^+], \quad k = 0, 1, \dots, N.$$

- Number of optimization variables: $(n + m)N + 1$
- The resulting discretized problem is a QP, possible solution methods are active set, interior point or projected gradient methods.